Question

1. Sample Java program to check whether the given number is prime palindrome or not.

2. Write a java program which reads marks of five subjects through command line and print the total and average.

Steps For Execution

Program 1:

1. Create a folder named PrimePalindrome.
2. Inside it, create a folder named record.
3. Save the Java program as PrimePalindrome.java inside the record folder.
4. Open the PrimePalindrome folder in IntelliJ IDEA/VS Code/Eclipse.
5. If there are no compilation errors, execute the program.
6. The program will display "Enter a number:".
7. Enter the required number.
8. The program will reverse the entered number and check whether it is a palindrome.
9. The program will check whether the entered number is prime by counting its factors.
10. If the number is both prime and palindrome, "Prime Palindrome" will be displayed.
11. Otherwise, "not a Prime Palindrome" will be displayed.
12. The program execution is completed.

Program 2:

1. Create a folder named FiveSubjMarks.
2. Inside it, create a folder named record
3. Save the Java program as FiveSubjMarks.java inside the record folder.
4. Open the FiveSubjMarks folder in IntelliJ IDEA/VS Code/Eclipse.
5. If there are no compilation errors, execute the program.
6. The program will display "Enter Mark 1:".
7. Enter the marks for the first subject.
8. Enter the marks for the remaining four subjects when prompted.
9. The program calculates the total marks by adding all five subject marks.
10. The program calculates the average by dividing the total marks by 5.

11. The total marks and average marks will be displayed.
12. The program execution is completed.

Algorithm 1:

Algorithm: Prime Palindrome
1. Start.
2. Read a number n from the user.
3. Initialize
   rev = 0
   temp = n
   count = 0
4. Reverse the number:
   a. Repeat while temp > 0.
   b. Find the last digit using temp % 10.
   c. Add the digit to rev.
   d. Remove the last digit using temp / 10.
5. Check whether the number is prime:
   a. Start a loop from i = 1 to n.
   b. If n % i == 0, increment count.
6. Check the conditions:
   a. If rev == n and count == 2,
      display "n is a Prime Palindrome".
   b. Otherwise, display "n is not a Prime Palindrome".
7. Stop.

Algorithm 2:

1. Start.
2. Read the marks of five subjects:

m1, m2, m3, m4, and m5.
3. Calculate the total marks: total = m1 + m2 + m3 + m4 + m5.
4. Calculate the average marks:
average = total / 5.0.
5. Display the total marks.
6. Display the average marks.
7. Stop.

UML Class Diagram

Program 1:

```
┌──────────────────────────────┐
|       PrimePalindrome        |
├──────────────────────────────┤
| + main(args : String[]) : void |
└──────────────────────────────┘
          |
          | uses
          ▼
┌──────────────────────────────┐
|          Scanner             |
├──────────────────────────────┤
|                              |
├──────────────────────────────┤
| + nextInt() : int            |
└──────────────────────────────┘
```

Program 2:

```
┌──────────────────────────────┐
|       FiveSubjMarks          |
├──────────────────────────────┤
|                              |
```

```
|  - sc : Scanner                 |
|  - m1 : int                     |
|  - m2 : int                     |
|  - m3 : int                     |
|  - m4 : int                     |
|  - m5 : int                     |
|  - total : int                  |
|  - average : double             |
├─────────────────────────────────┤
|                                 |
|  + main(args : String[]) : void |
└────────────────┬────────────────┘
        |
        | uses
        ▼
┌─────────────────────────────────┐
|          Scanner                |
├─────────────────────────────────┤
|                                 |
├─────────────────────────────────┤
|  + nextInt() : int              |
: └───────────────────────────────┘
```

Code

Program 1:

```
package javacore;
import  java.util.Scanner;
public class PrimePalindrome {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter a number: ");
      int n = sc.nextInt();
```

```
        int rev = 0, temp = n, count = 0;

        // Check palindrome
        while (temp > 0) {
            rev = rev * 10 + temp % 10;
            temp = temp / 10;
        }

        // Check prime
        for (int i = 1; i <= n; i++) {
            if (n % i == 0)
                count++;
        }

        if (rev == n && count == 2)
            System.out.println(n + " is a Prime Palindrome");
        else
            System.out.println(n + " is not a Prime Palindrome");
    }
}
```

Program 2:

```
package javacore;

import java.util.Scanner;

public class FiveSubjMarks {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Mark 1: ");
        int m1 = sc.nextInt();

        System.out.print("Enter Mark 2: ");
        int m2 = sc.nextInt();

        System.out.print("Enter Mark 3: ");
        int m3 = sc.nextInt();

        System.out.print("Enter Mark 4: ");
        int m4 = sc.nextInt();
```

```
        System.out.print("Enter Mark 5: ");
        int m5 = sc.nextInt();

        int total = m1 + m2 + m3 + m4 + m5;
        double average = total / 5.0;

        System.out.println("Total = " + total);
        System.out.println("Average = " + average);
    }
}
```

Output:

Program 1:

Enter a number: 131

131 is a Prime Palindrome

Program 2:

Enter Mark 1: 80

Enter Mark 2: 75

Enter Mark 3: 90

Enter Mark 4: 85

Enter Mark 5: 70

Total = 400

Average = 80.0